英飞凌AIROC™无线产品助力AI未来

-- 赋能边缘，无“线”可能

Barry Weng

March 2025

public

议程

1 AI边缘计算对无线网络的迫切需求与核心挑战 3

2 英飞凌AIROC™ Wi-Fi 6 产品 —— 智联万物，赋能边缘 7

3 AI边缘实战——英飞凌无线产品的场景化优势 11

4 英飞凌AIROC™ Wi-Fi 6 MCU —— 连接万物，智在边缘 18

5 总结与展望——AI边缘计算的无线智联未来 25

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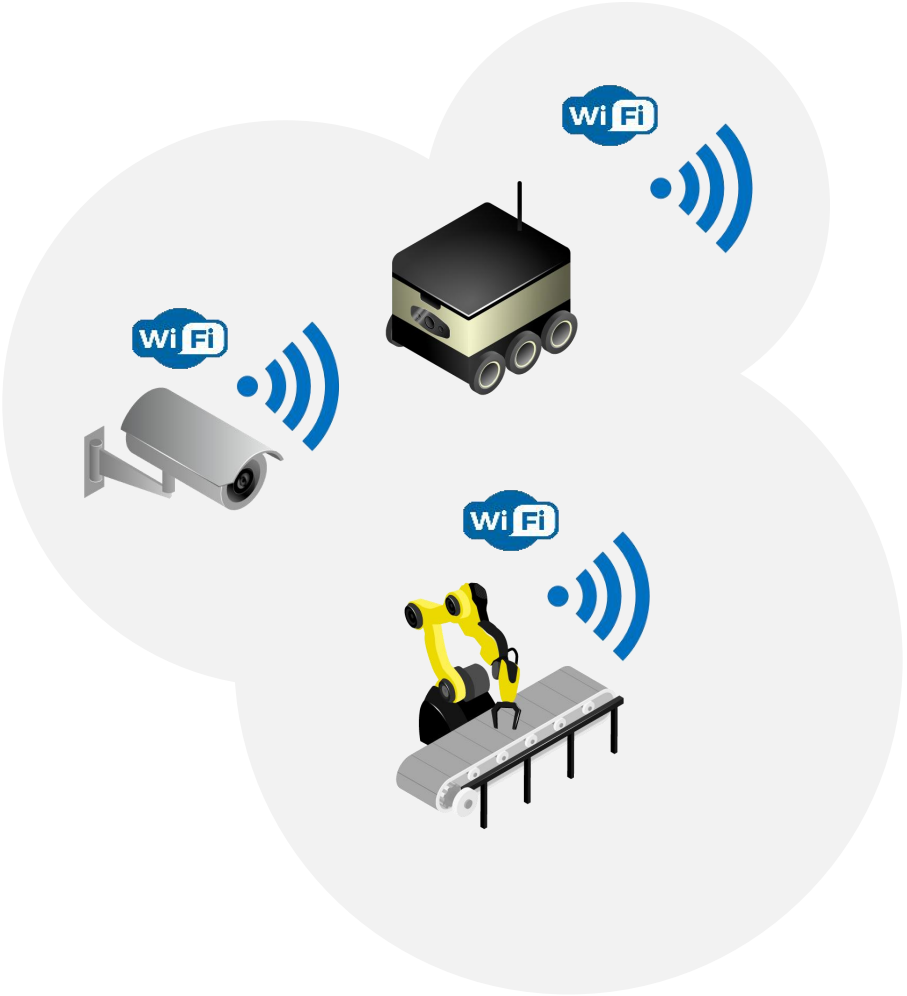
AI边缘计算：无线网络的新需求与新使命

智能边缘设备



更新环境

部署AI模型并优化任务控制

云端的设备管理

实时数据传输：生成准确的影像

低时延控制：任务控制

高效数据回传：迭代AI训练

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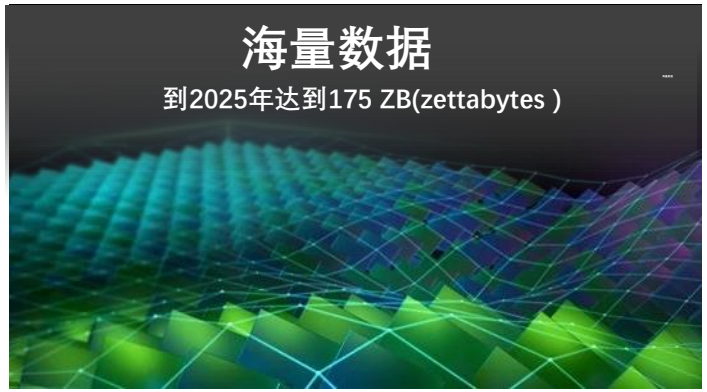
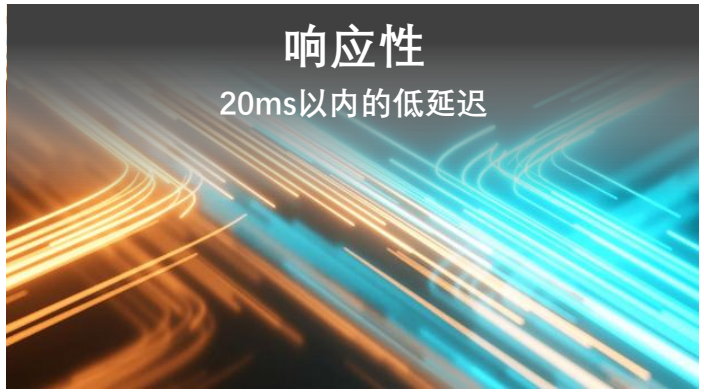
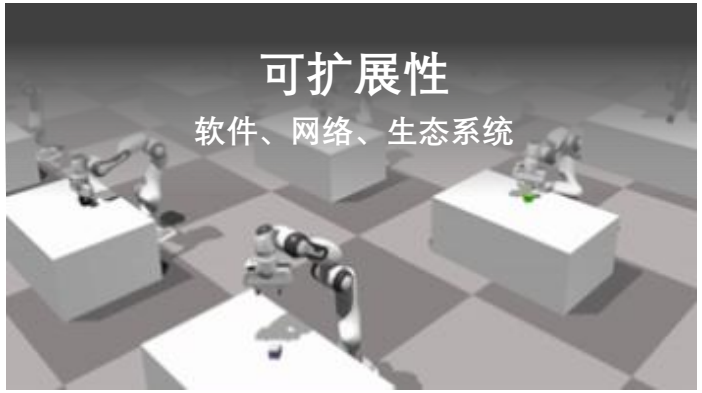
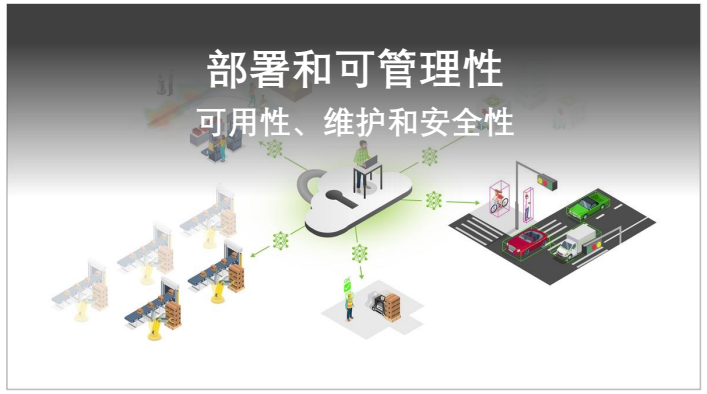
无线技术落地AI边缘计算的五大核心挑战









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从挑战到突破：无线网络如何支撑边缘AI计算

边缘AI计算面临的挑战	英飞凌Wi-Fi 6无线的特性
海量数据 2025年的175 ZB(zettabytes)	高吞吐量
移动性 能效和紧凑尺寸	低延迟 强健的网络
响应性 20ms以内的低延迟	低功耗 扩展范围
可扩展性 软件、网络和生态系统	使用AI验证软件
部署和可管理性 可用性/维护/安全性	多层安全

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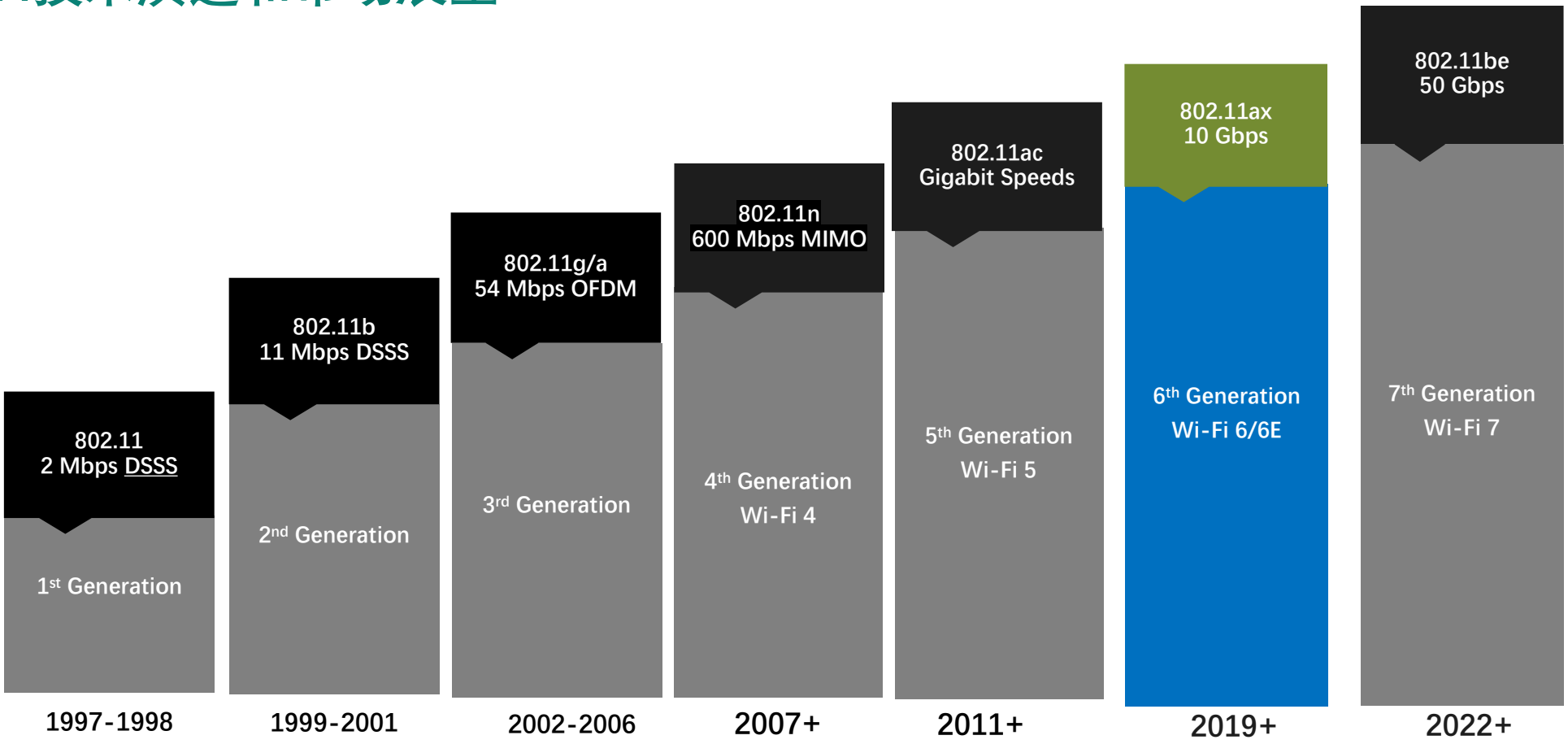
最新Wi-Fi技术演进和市场展望



可靠	速度,安全	连接效率和用户体验

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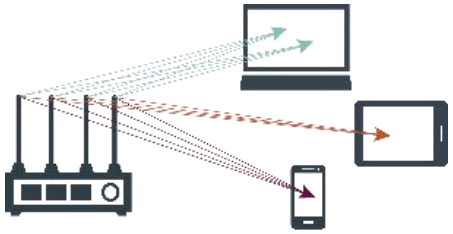
Wi-Fi 6/6E带来的新功能

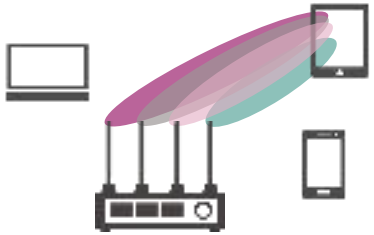
Wi-Fi6/6E技术明显的改善了**网络连接的效率**和**终端用户的体验**

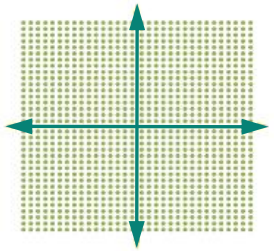
Multi-User MIMO



OFDMA

Beamforming

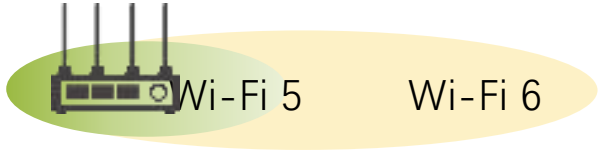




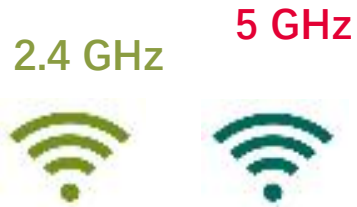
1024-QAM

Wi Fi 6™ CERTIFIED

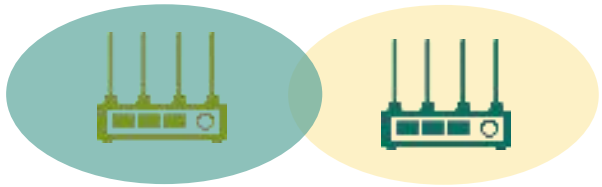
Increased Symbol Time



Spatial Streams



BSS Coloring



Target Wake Time

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英飞凌AIROC™ Wi-Fi 6/6E 产品组合

Wi-Fi/BT Combo			Wi-Fi only	Wi-Fi/BT MCU
55573/2	55571/0	55513/2/1	55533/2/1	55913/2/1
Wi-Fi 2x2 80MHz	Wi-Fi 1x1 80MHz	Wi-Fi 1x1 20MHz	Wi-Fi 1x1 20MHz	Wi-Fi 1x1 20MHz
Tri-band / Dual-band 2.4/5 GHz	Tri-band / Dual-band 2.4/5/6 GHz	Tri-band / Dual-band 2.4/5/6 GHz	Tri-band / Dual-band 2.4/5/6 GHz	Tri-band / Dual-band 2.4/5/6 GHz
Max. Phy Tput: 1200Mbps	Max. Phy Tput: 600Mbps	Max. Phy Tput: 143Mbps	Max. Phy Tput: 143Mbps	Max. Phy Tput: 143Mbps
PCIe, SDIO	PCIe, SDIO	SDIO/GSPI	USB/ SDIO/GSPI	SDIO/GSPI
Hosted Wi-Fi	Hosted Wi-Fi	Hosted Wi-Fi	Hosted Wi-Fi	Wi-Fi Connected MCU (CM33 192MHz)
Bluetooth 5.3 (Hosted BT/LE Audio)	Bluetooth 5.3 (Hosted BT/ LE audio)	Bluetooth 5.4 (Embedded BT/LE Audio)	-	Bluetooth LE 5.4
FCBGA, WLCSP, WLBGA	FCBGA, WLCSP, WLBGA	5551x: 0.35mm WLBGA 6L HDI PCB 5554x: 0.4mm WLBGA 4L HDI PCB	WLBGA 0.4mm	WLBGA 0.35mm
已量产	已量产	已量产	预计2025年1季度量产	已量产

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AIROC™ CYW5557x系列：独特的差异化

CYW5557x系列

2x2用于高性能
1x1用于低功率

- **Release 1和Release 2特性**
- **三波段：**2.4GHz、5GHz、6GHz
- **MU-MIMO, OFDMA, 1024QAM**
- 1.2Gbps max. PHY数据速率
- **软AP 和 Station modes**
- **蓝牙 5.3:** LE音频， Auracast
- 工业温度：-40 到85℃

英飞凌增强

- 网络鲁棒性
- 范围展延
- 低功耗
- Wi-Fi/蓝牙®的智能共存
- 增强的安全性和保护

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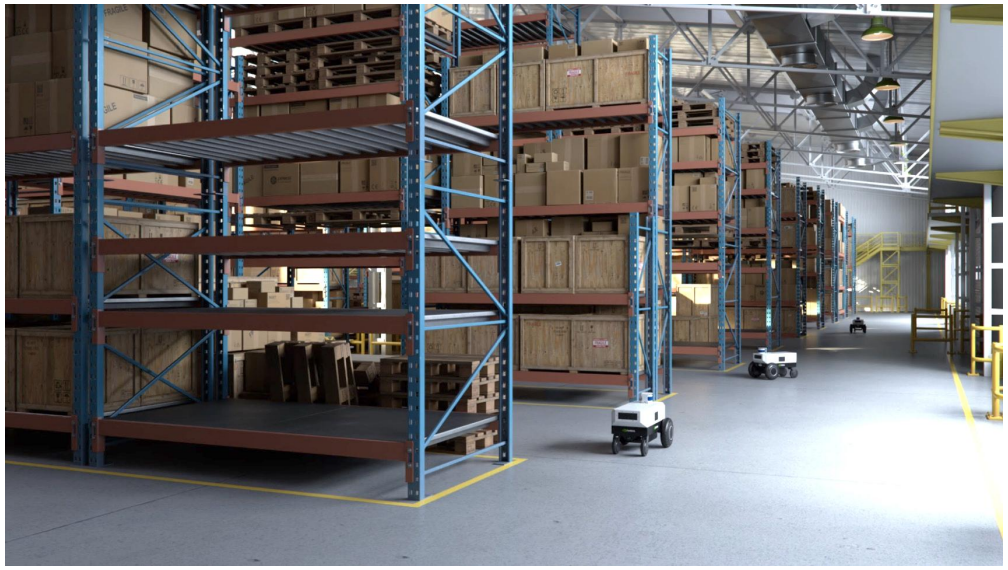
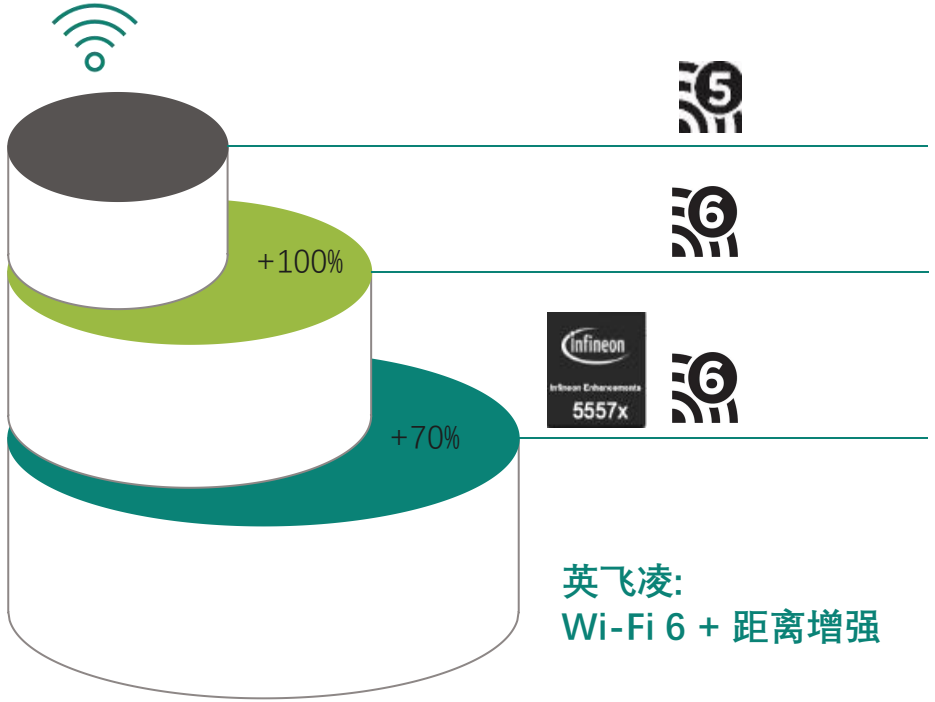
CYW5557x 范围延长:
AMR（自主移动机器人）车队管理需要更广的覆盖范围



挑战:
- 机器人在大型仓库中作业
- 通信因通过障碍物导致信号衰减
- 机器人失去连接/通信
- 高效电源延长服务时间

英飞凌Wi-Fi 6+距离增强

- 与Wi-Fi 5产品相比，范围提升了2倍
- 与Wi-Fi 6产品相比，范围提升了70%



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CYW5557x 低功耗：
从硬件到应用层的低功耗功能

Wi-Fi芯片级节能增强

在接收模式下使用先进的低功耗特性，最高可省电20%

在射频校准过程中最多可省电30%

通过动态改变空间流和带宽延长电池寿命

目标唤醒时间：根据不同用例需求，量身定制的灵活配置

先进技术过滤掉不相关的Wi-Fi数据包，节省时间、降低功耗

通过网络层卸载实现系统节能

网络offload特性允许主机处理器休眠更长时间

LE唤醒特性通过允许主机和Wi-Fi子系统休眠来省电

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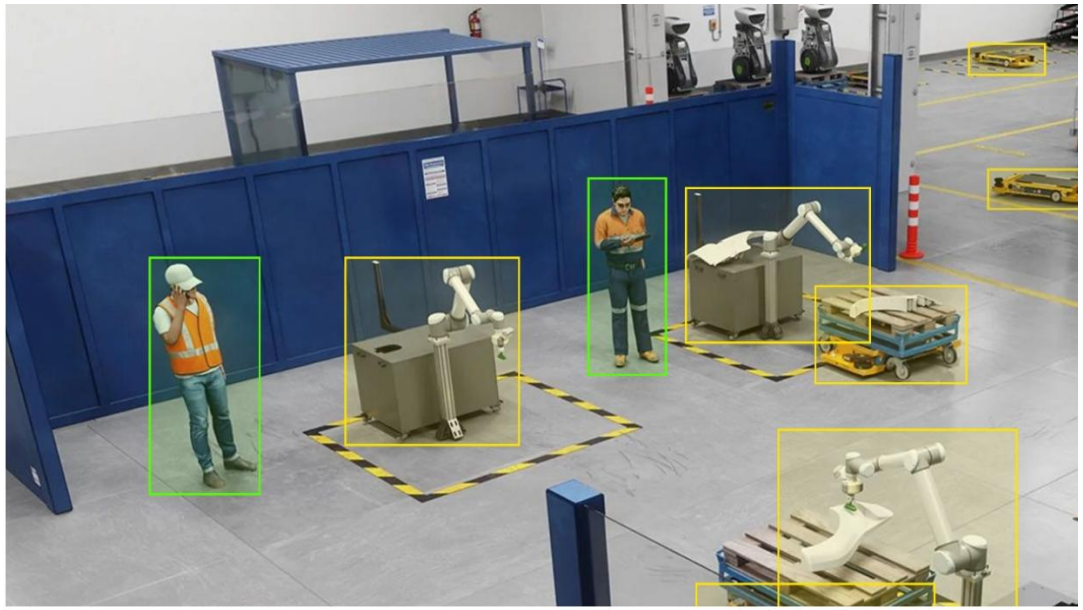
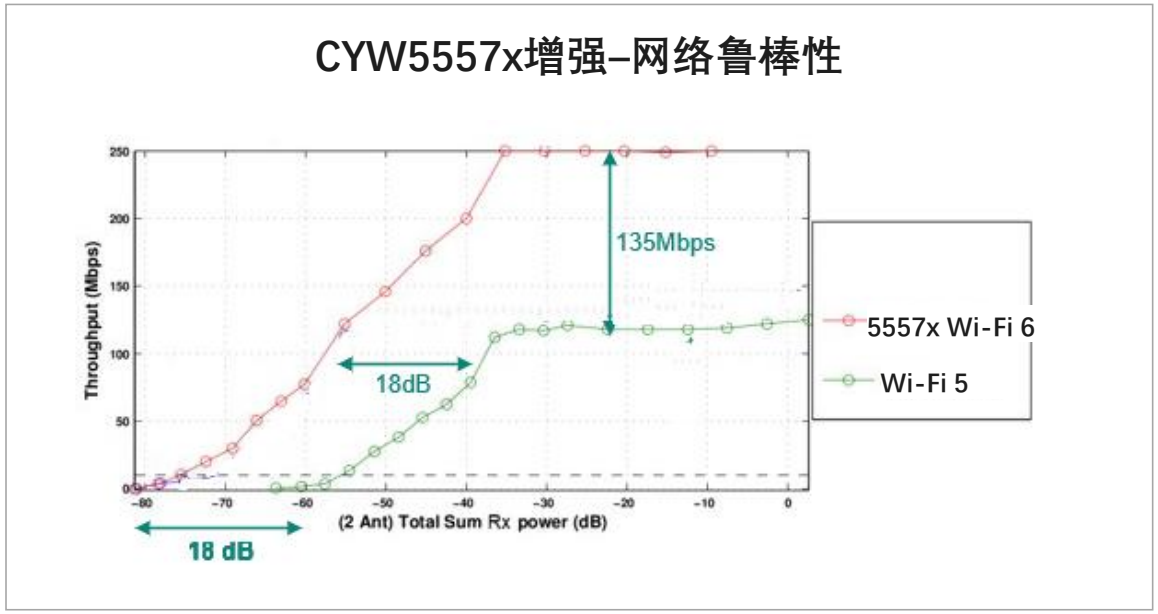
CYW5557x 网络鲁棒性:
基于视觉的远程操作要求低延迟





挑战:

- 摄像头处于嘈杂的环境中
- 摄像头在偏远区域
- 防止机器人和工人碰撞的低延迟

在相邻信道干扰情况下:

- CYW5557x以比Wi-Fi 5设备更远的距离，接收相同吞吐量的信号（18db灵敏度差异）
- CYW5557x在相同距离，提供比Wi-Fi 5设备更高的吞吐量

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CYW5557x 低延时:
基于视觉的远程操作要求低延迟

挑战：

- 远程通信和操作
  - 多摄像头实时视频流
  - 需要精确的反馈

从摄像机捕捉到动作要求:

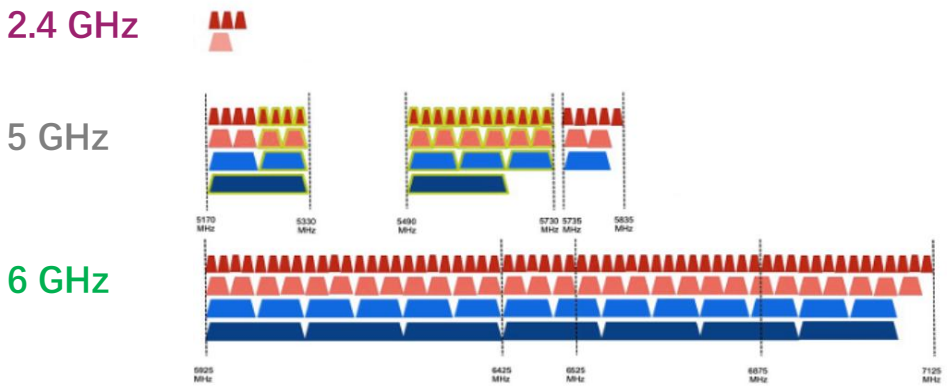
- 更快的视频处理实现精确控制
- 低延迟和实时Wi-Fi，最大限度地减少运动延迟

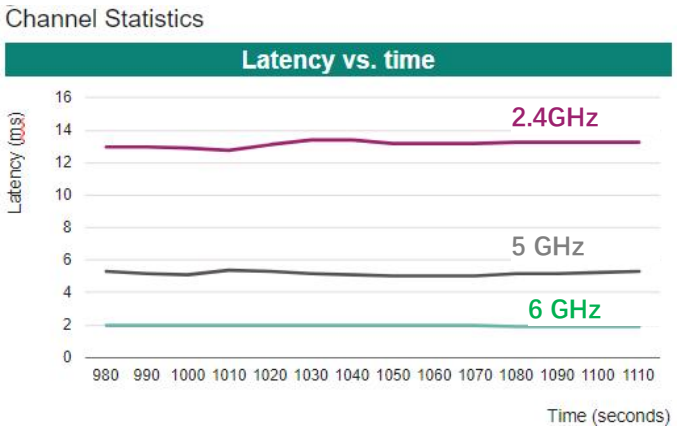
6GHz具有如下优势：

带宽是2.4GHz的20倍

20MHz信道比2.4GHz多59倍

延迟为2.4Ghz的15%





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在 Linux、Android 和 RTOS 上移植和优化的软件

AIROC™ Wi-Fi Combos –产品组合和软件支持

AIROC™: Leading Wi-Fi portfolio for IoT and Automotive…

Wi-Fi 6/6E

- **5557x** — **Industry's best range Wi-Fi 6/6E + LE-Audio**
  >40% better than standard Wi-Fi 6/6E
- **5551x** — **IoT Optimized Wi-Fi 6/6E + BT/BLE-Audio**
  >40% better than standard Wi-Fi 6
- **5591x** — **Low Power Wi-Fi 6/6E + LE Connected MCU**
  Best range performance and rich peripherals

Wi-Fi 4/5

- **5459x** — **Reconfigurable 2x2 MIMO and RSDB**
  High-performance, whole home coverage
- **4373** — **Best general-purpose Combo**
  Industrial temp. qualified, USB/PCIe/SDIO
- **43022** — **Industry's lowest power**
  >70% battery life vs. closest competitor
- **43439** — **Best entry-level Wi-Fi/BT Combo**
  Family shipped over 300Mu

Automotive versions available

…with software ported on and optimized for…

Linux/Android	RTOS
General purpose processing	**General purpose processing**
NVIDIA — Jetson, Xavier, Orin	Infineon — Infineon PSOC6, PSOC Edge
STM — STM32 MP15x/13x, **MP23x, MP25x**	STM — STM32H7, H5, U5, L5, N6
D-Robotics — RDK X3, RDK X5	RaspberryPi — PICO-W
Rockchip — PX30, RK3308, RK3399, RK3566, **RK3568, RK3588**	更多平台正在增加中…
RaspberryPi	

目前支持多达 14 个平台
（签署NDA后开放）

新平台以绿色文本标记

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平均家庭 Wi-Fi 使用量正在迅速增长 – 每个家庭将有 50 多台连接的设备

Work / Personal
- Laptops/PCs
- Tablets
- Printer
- Smartphones
- Smartwatches
- Smartglasses

Entertainment
- TVs
- Streaming sticks
- Audio
- Gaming
- AR/VR

Appliances
- Refrigerators
- Dish washers
- Coffee machines
- Air conditioning
- Washers
- Cooktops
- Dryers

Home Automation
- Smart Speakers
- Garage door openers
- Door locks
- Doorbells
- Chimes
- Thermostats
- Temperature sensors
- Smoke & detectors
- Security cameras
- Lighting
- Cleaning robots
- Sprinkler controllers
- Other sensors

Others
- Exercise machines
- Baby monitors
- Toys
- Grills
- EV chargers
- Solar inverters
- Air purifiers
- Utility meters

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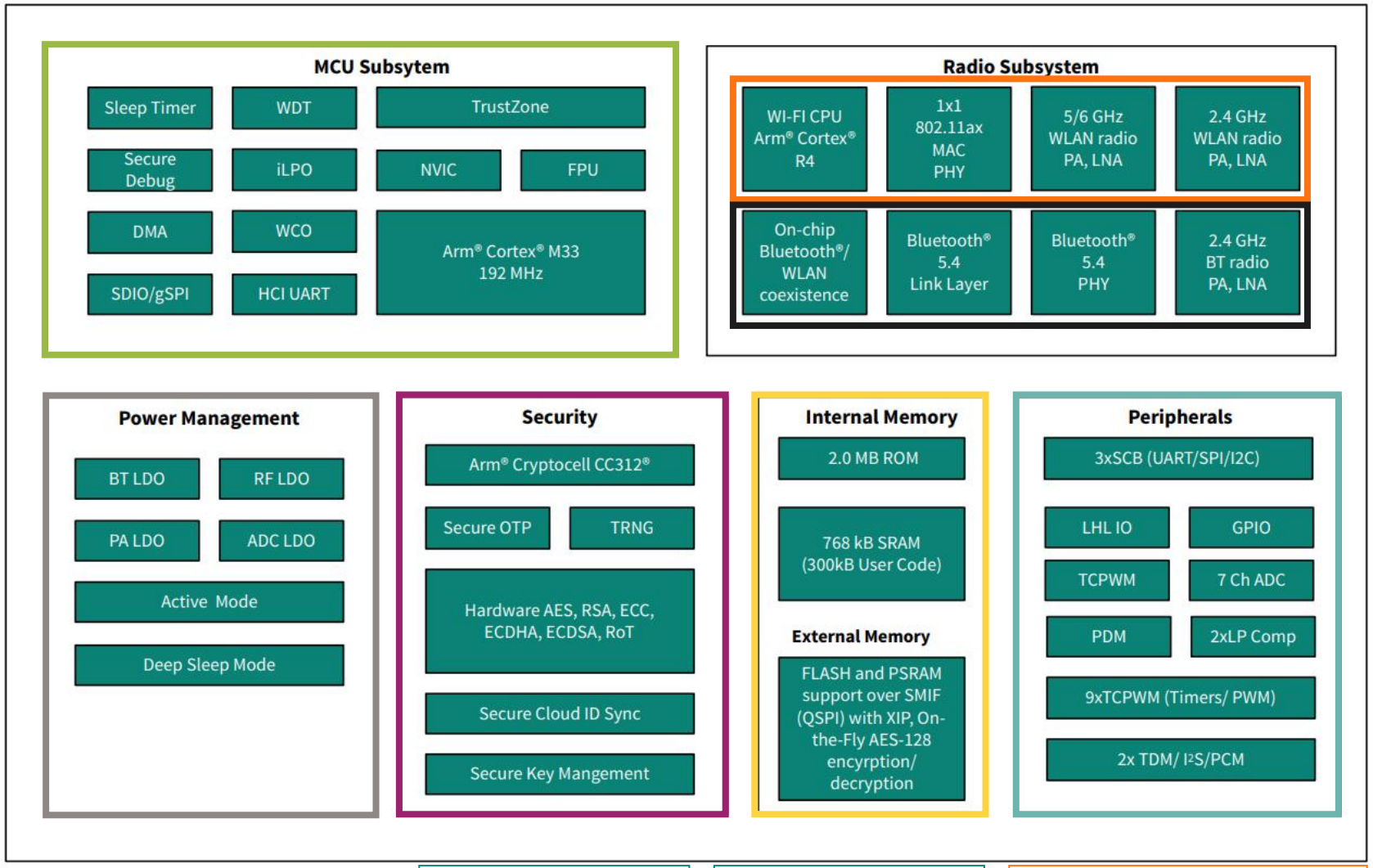
CYW5591x无线MCU系列打造一流的物联网产品

High-Performance
192MHz
Arm Cortex M33

Integrated PMU with Active and Deep Sleep Modes



Long-Range
Low-Power Wi-Fi 6/6E
Tx Power up to +24dBm
Rx Sens -101.5dBm
802.11ax + IFX Range Improvements

Dedicated BLE
Low-Power BLE 5.4
Tx Power +19/+13/+4dBm
LELR-S8: -111.5dBm

Extensive
Peripheral Support

Display drive capable over SPI to 24 MHz

PSA Level 2 Certifiable

External FLASH/PSRAM up to 16MB

Expand memory for data storage over SPI

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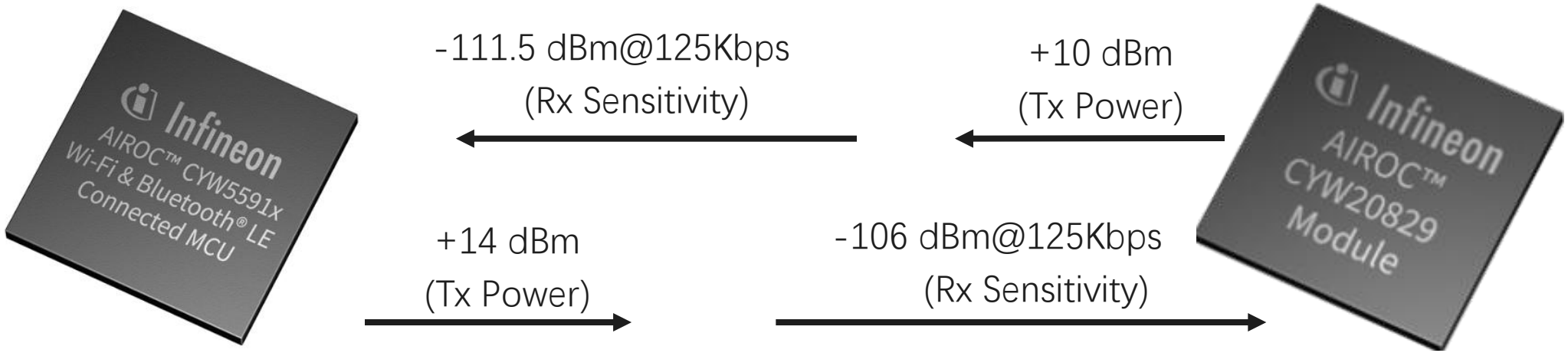
CYW5591x: BLE长距离功能

配合CY20829提供最好组合



Link Budget

BT Up-Link (Node To Gateway): 121.5 dB

BT Down-Link (Gateway To Node): 120 dB

Range @ 120 dBm Link Budget

户外: ~3 公里

室内 ~400米

* Range is Calculated theoretically based on Range Calculator (https://wiki.lm-technologies.com/wireless-link-budget-calculator/)

CYW5591x 的灵敏度比同类产品高 9dB，大大扩展了范围

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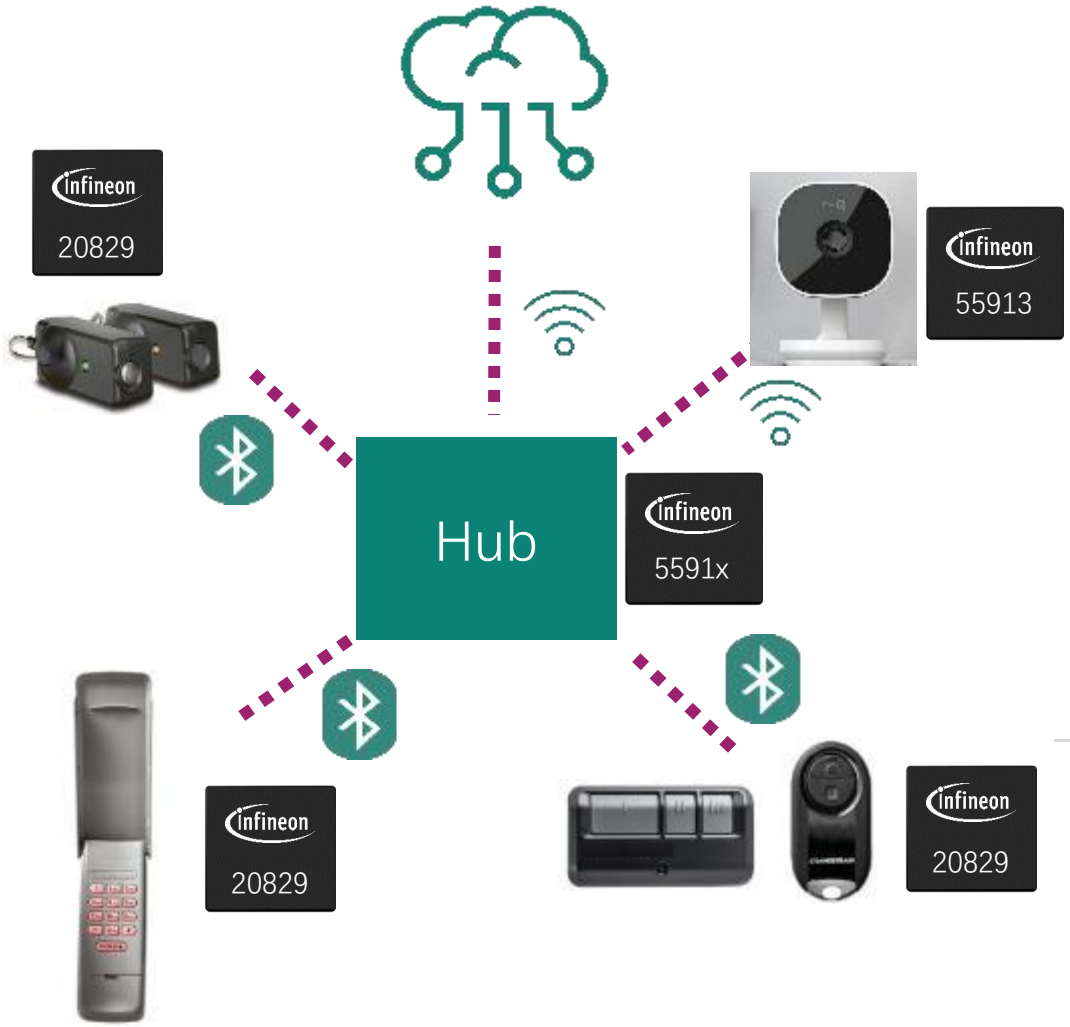
CYW5591x：Wi-Fi + BLE 网关Hub系统方案

优化和可管理的低功耗、安全、智能访问网络



5.4 Wi-Fi 6(E)

20829 5591x

More range and robustness than ever

Future proof your design

Up to 30% lower system cost due to highly integrated MCU

Longer battery life with low power connection

1 msec latency without a dongle

适用于广泛的应用

- 工业
- 智能家居
- 穿戴设备
- 智能楼宇
- 智能传感器

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英飞凌 W-Fi 6 MCU + 毫米波雷达

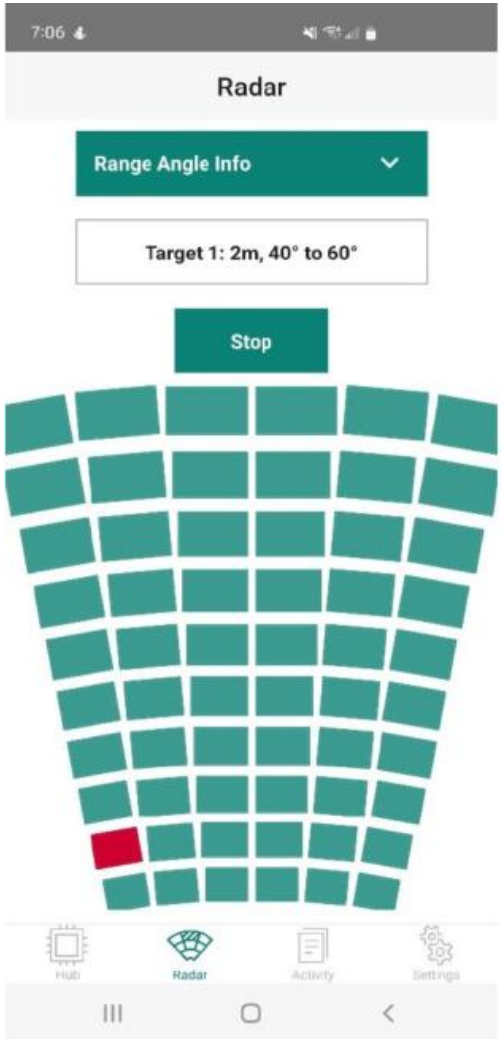
存在和分区检测解决方案

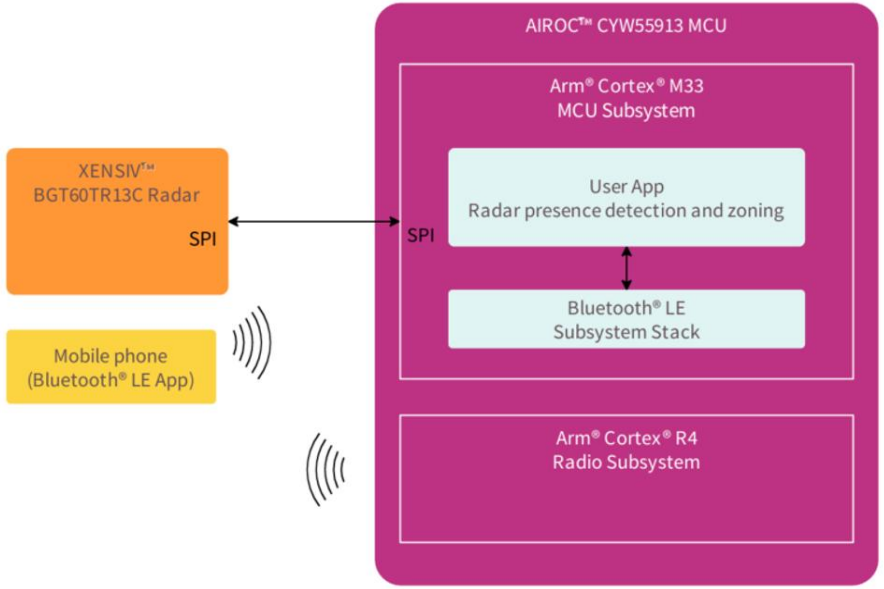
› **英飞凌解决方案**

- 产品： BGT60TR13C + CYW55913
- Wi-Fi 软件 offloading与运行在Wi-Fi MCU上的算法

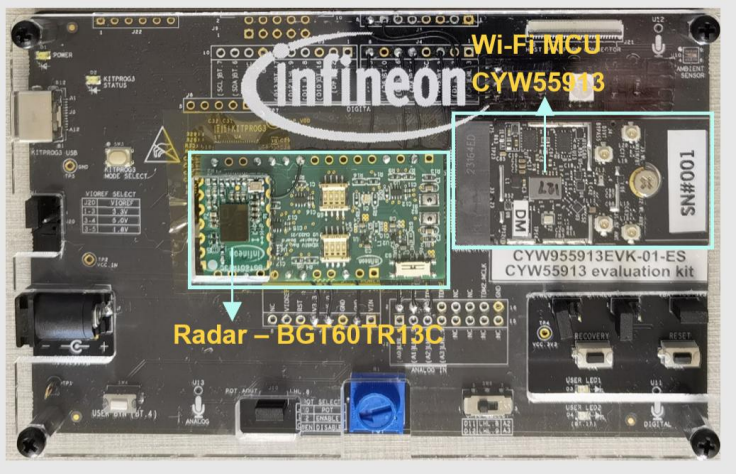
› **带给客户的价值**

- 降低BOM
- 芯片集成度 Wi-Fi 6 MCU:CYW5591x
- 减少假阳性（PIR的 弱点）
- Radar 的高阶功能使用案例
- 简化系统架构





系统框架



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英飞凌用于Wi-Fi感知的Wi-Fi CSI（信道状态信息）

2018
CSI for Wi-Fi 5
43012

2024
CSI for Wi-Fi 6
5557x/5551x/5591x

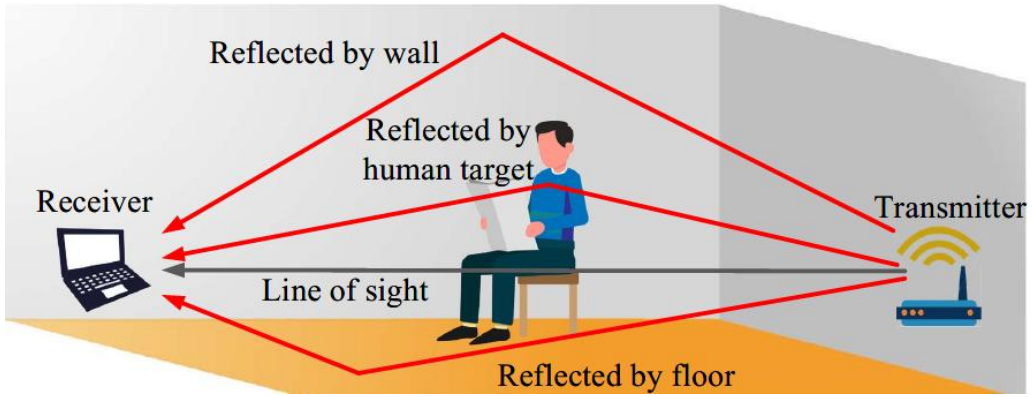
2026
CSI for Wi-Fi 7 (11bf)
Standardized CSI

Wi-Fi感知的价值

- 增值：为现有Wi-Fi产品增加价值
- 成本：利用现有Wi-Fi,不增加额外器件
- 升级：基于软件，可通过OTA不断优化
- 伙伴：合作伙伴提供AI/ML算法

典型场景

- 存在检测
- 运动检测



Wi-Fi CSI

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总结与展望

- AI边缘计算的新需求 Wi-Fi
  - 无线网络需满足超低时延、超高带宽、超高密度**三大需求**
  - Wi-Fi 6/6E技术已成为AI边缘的核心支撑
- 英飞凌AIROC™ Wi-Fi 6 产品线
  - 覆盖2x2 80MHz的高性能市场，1x1 20MHz低功耗市场。
  - 独特的优势:更长的距离，更Combo好的鲁棒性，更省电，更安全。
  - 运用场景和生态: 模组，平台支持，软件和技术支持等。
- 无线在AI边缘场景实战
  - 远程视觉操作和摄像头抗干扰场景中，英飞凌方案显著提升性能
  - 6GHz频段和BSS Coloring技术解决复杂环境干扰问题
- 英飞凌AIROC™ Wi-Fi 6 MCU的边缘价值
  - 省电，安全，长距离连接。
  - 与英飞凌MCU和传感器等产品组合和更多应用解决方案。

赋能边缘，无“线”可能

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